Revealing SARS-CoV-2 protein architectures and function by in situ MS proteomics and integrating modeling

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Motivation – Gap to fill

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SARS-CoV-2 encodes **29 major proteins** - 16 non-structural proteins (Nsp1-16), 4 structural proteins (S, E, M, N), 9 major Orfs.

- Gaps in the knowledge of structure and function apply to several viral proteins. Current purification schemes may cause proteins to denature, aggregate or disassemble.
- To overcome such artifacts and to gain detailed protein structural information , **in situ techniques** are required.

Targeted In situ CLMS protocol

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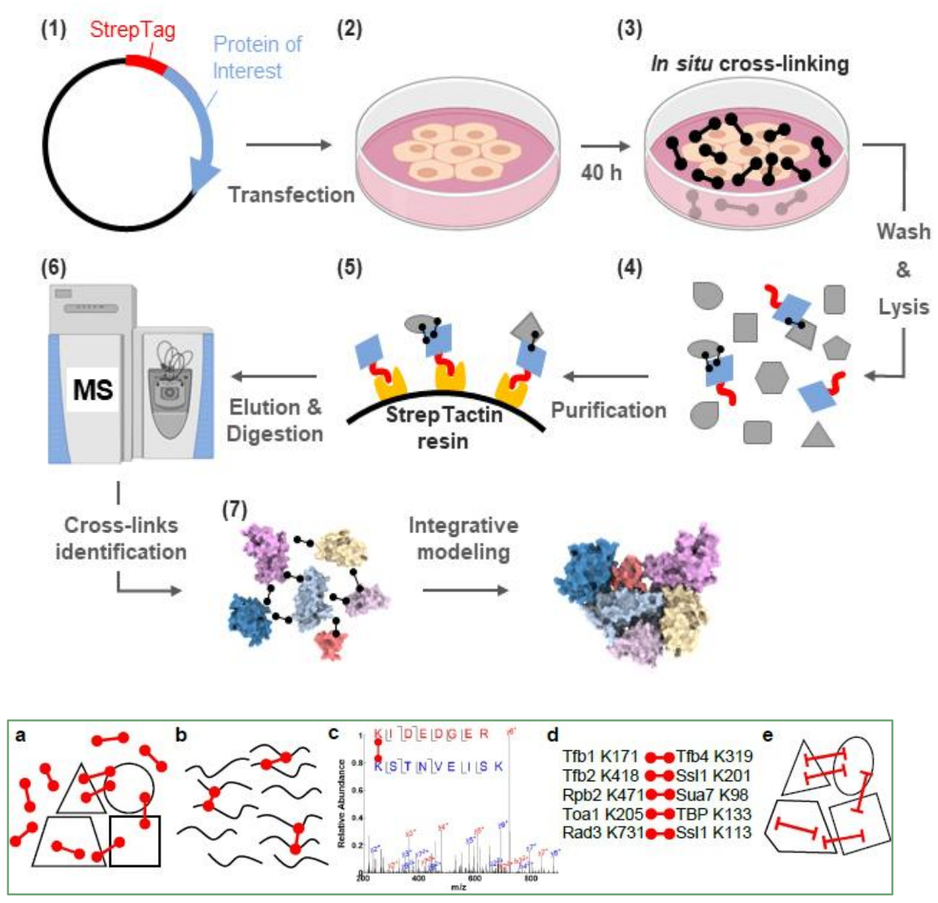
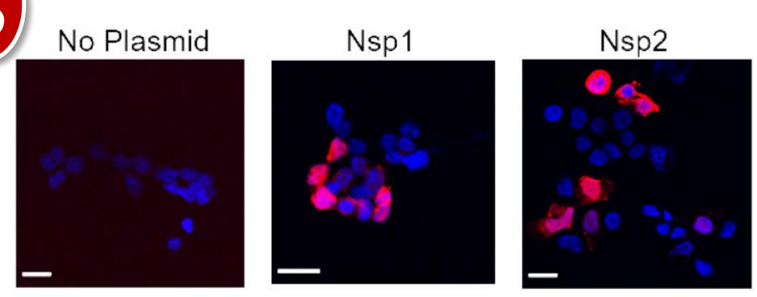
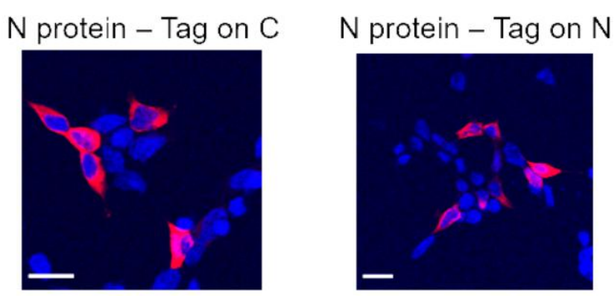


No aggregates in expressed cells

3





Testing aggregation in transfected cells. No signs of aggregation. Pink - Staining against the Strep-tag. Blue - DAPI staining. Expression of the N' and C' tag on N protein is identical. Scale bar - 20 µm

Nsp2 – structural architecture based on in situ CLMS

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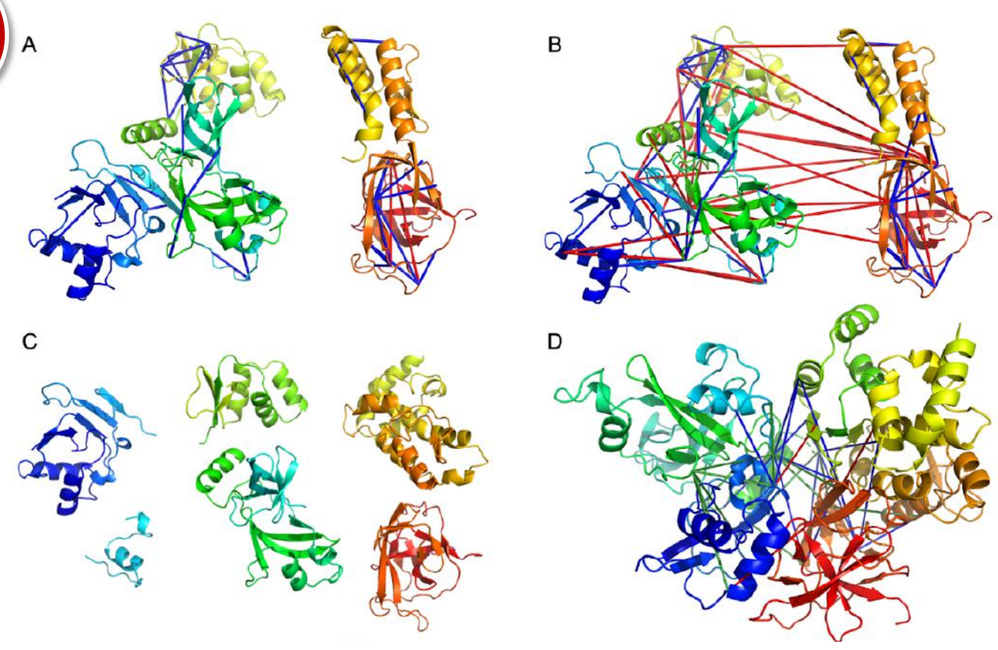

Integrative modeling of Nsp2. The modeling is based on cross-links (satisfied CX are in blue, unsatisfied in red). **(A)** Intra-domain CX, **(B)** All CX mapped onto the AlphaFold2 model. **(C)** Six domains as input to modeling pipeline. **(D)** Representative result of the integrative modeling with inter-domain CX (red - violated cross-links).

N-protein –structure, evolution and dynamics

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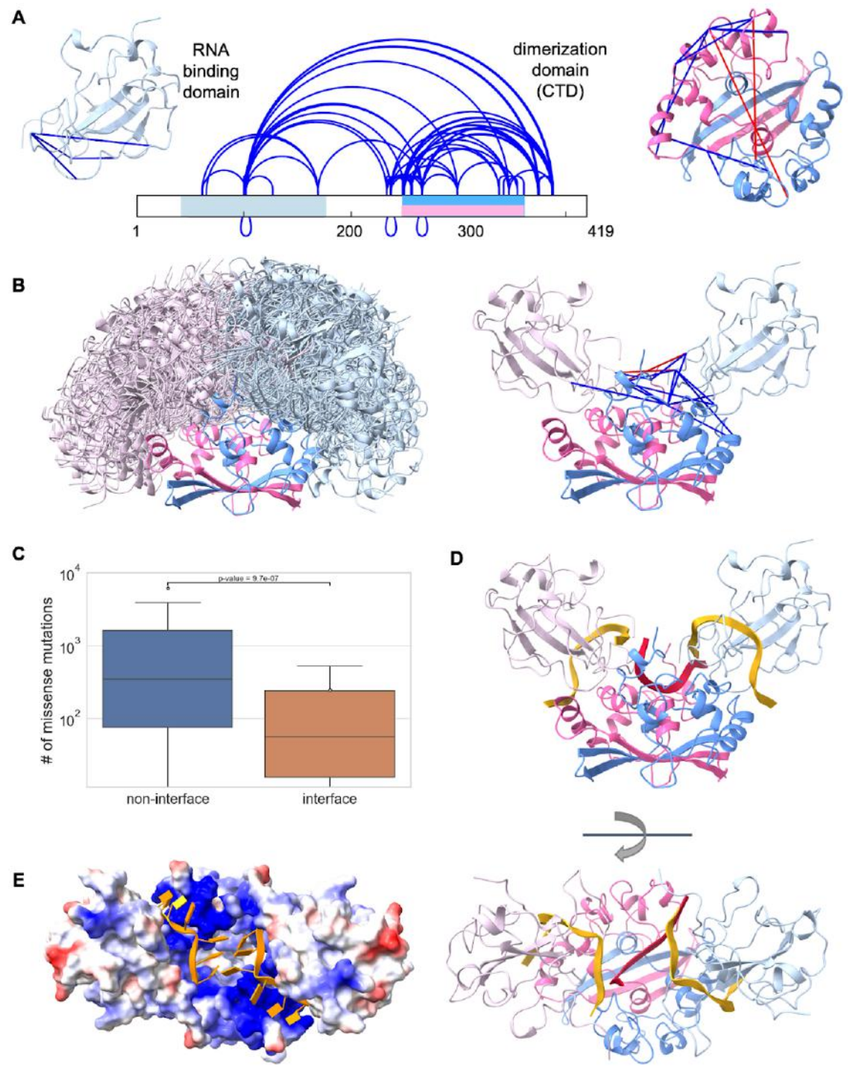

Integrative modeling of the N protein. (A) In situ CX identified within the N protein. The domains are marked in light blue (RNA binding domain) and pink/blue (dimerization domain). The intra-domain CX (satisfied CX are in blue, unsatisfied in red) are mapped onto available structures (PDBs 7act and 6zco) **(B)** Best scoring docking models. **(C)** Missense mutations observed in interface and non-interface residues. **(D)** RNA fragments bound to dimerization domain (red) and RNA binding domains (gold). **(E)** The electrostatics potential with two RNA fragments (gold).

Nsp1 – Blocking mRNAs entry on the 40S ribosome

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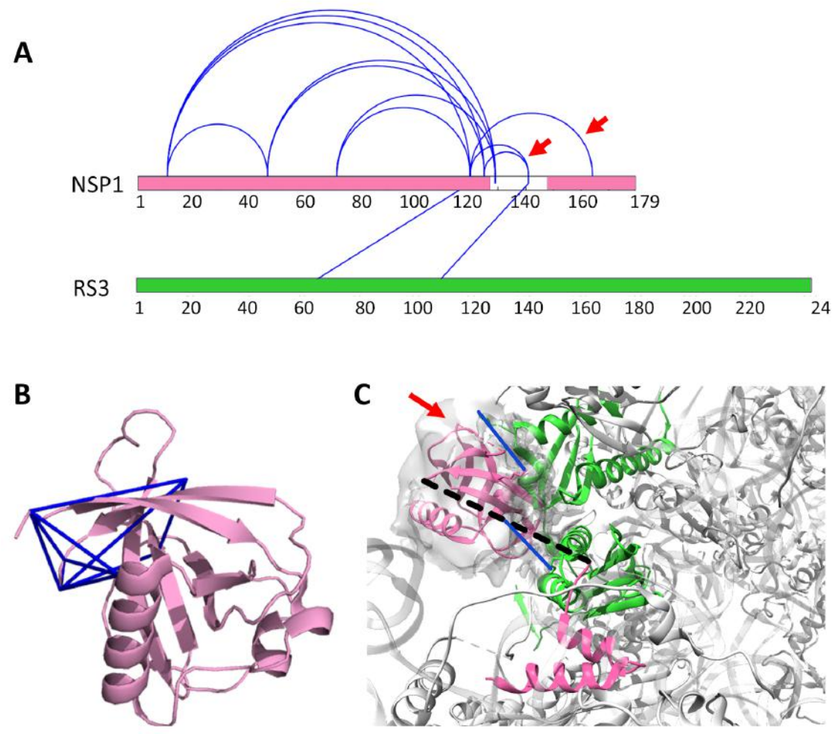

Integrative modeling of Nsp1. (A) In situ CX identified within and between Nsp1 and ribosomal protein (RS3). **(B)** In situ CX (blue) within Nsp1 mapped to PDB 7k3n. **(C)** Mapping of the Nsp1-RS3 crosslinks (blue) onto the cryo-EM structure of the C-terminal of Nsp1 (pink) bound to the 40S ribosomal subunit (gray) (PDB: 6lzw).

Conclusions

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- **In-vitro affinity purification** is prone to non-native interactions
- Cross-linking captures **"native" 3D structures and PPI**
- CLMS & Integrative modeling provide **atomic resolution**

- **Nsp1 -** blocks mRNA entry by binding to Ribosomal S3 (RS3)
- **N –** Dimer, strong RNA binding domain, condensate
- **Nsp2** – Zinc binding protein, much more compact protein. 3D conserve

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